ADEEL AHMAD

Undergraduate Researcher in Network Security, Federated Learning, and Vehicular Networks
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SUMMARY

Undergraduate researcher in network security for wireless and vehicular networks. Research covers poisoning attacks on federated learning in Open RAN, privacy leakage in vehicular federated learning, and explainable DDoS detection in software-defined networks. First author of a journal paper published in IEEE Access and another under second-round review at Elsevier Computer Networks, and co-author of a paper accepted at IEEE VTC2026-Fall. Developer of OmniWalk, an offline AI white cane presented at the KSCDR AI Hackathon for People with Disabilities and selected for the SAIF Riyadh Hackathon. Releases open code with every first-author paper. Seeking a funded Master's or PhD position in network security or vehicular networks.

EDUCATION

Bachelor of Information Technology, Islamic University of Madinah, Saudi Arabia — 2023 – Present
- CGPA 4.0 / 5.0

Higher Secondary Education, Punjab College of Commerce, Pakistan — 2020 | 92%

Secondary Education, Govt. High School Qila Didar Singh, Pakistan — 2017 | 90%

HONOURS AND AWARDS

Fully Funded, Merit-Based Saudi Government Scholarship, Islamic University of Madinah — 2023 – Present

AI Hackathon for People with Disabilities, King Salman Center for Disability Research (KSCDR), Saudi Arabia — 2026
Selected participant; presented the initial working prototype of OmniWalk.

Security and Innovation Fair (SAIF) Hackathon, Riyadh, Saudi Arabia — 2026
Selected participant; shortlisted to compete with our project.

PUBLICATIONS

Google Scholar: scholar.google.com/citations?user=EjQm_kcAAAAJ

Published & Accepted

XAI-SDN: An Explainable Entropy-Guided Machine Learning Framework for Real-Time DDoS Detection in Software Defined Networks
IEEE Access | First author | Published, 2026 | IEEE Xplore: ieeexplore.ieee.org/document/11741196
Code: github.com/adeliusa486/XAI-SDN
A DDoS attack can take down the services people rely on, from banking to hospital systems. Most detectors give a yes-or-no answer that operators cannot check. XAI-SDN detects attacks in under a millisecond and shows why each alert was raised, so a person stays in control of the decision. We tested it on 3.59 million real network flows and in a live Mininet network.
Skills: Software-defined networking, Mininet, Random Forest, SHAP explainable AI, real-time detection

IEEE 104th Vehicular Technology Conference (VTC2026-Fall)
Conference paper | Co-author | Accepted, 2026
Federated learning is meant to keep drivers' data inside the car. We showed that the updates a car shares can still reveal which car sent them, a tracking risk drivers would never notice, and we tested defenses against it.
Skills: Federated learning, PyTorch, privacy attacks and defenses, vehicular networks

Under review

Which Robust Aggregation Rule Should You Trust for Open-RAN? An Empirical Study of Federated Poisoning Defenses on Recorded Cellular Measurements
Computer Networks, Elsevier | First author | Under review, second revision round
Code: github.com/adeliusa486/oran-fed-robust
Mobile networks are starting to train AI across many cell towers, and one hacked tower can poison the model for everyone. Existing defenses had mostly been tested on image data. We tested 13 defenses against 7 attacks on real 4G and 5G data in 4,290 runs. None survived every attack, and two popular ones did worse than no defense. We also proposed a new defense that fixes one of these failures, which gives operators real evidence when they protect networks that carry emergency calls.
Skills: Federated learning, Byzantine-robust aggregation, PyTorch, 4G/5G data analysis, significance testing

Research and Work Experience

Research Assistant, Islamic University of Madinah 2025 – Present
Faculty of Computer and Information Systems

- Led first-author studies in federated learning security and network security, from the research question to the published and submitted papers.
- Design experiments, write and test code, run large evaluations, and draft manuscripts and reviewer responses.
- Release the code, configurations, and per-seed results of each paper so others can check our numbers.

Intern, Asaqefah Business Development, Saudi Arabia Jun 2026 – Aug 2026

- Built an MVP of a multilingual app that guides visitors at Al-Masjid an-Nabawi, with an AWS operations dashboard for crowd management. Owned the full stack, from real-time navigation to cloud architecture.

Projects

OmniWalk: Offline AI White Cane for Blind and Low-Vision Pedestrians Graduation project, 2026 – Present
Industrial prototype: adeliusa486.github.io/OmniWalk

- Tells the user what is ahead, in which direction, and how far away through earbuds, and warns of obstacles, kerbs, and drop-offs through vibration and speech.
- Runs fully on the cane with no internet, phone, or subscription. A separate safety controller keeps obstacle alerts working even if the main computer fails.
- Built on lessons from our network research: AI on the device, testing in real conditions, and design for failure.

Tech: Raspberry Pi 5, Hailo-8L AI accelerator, YOLO object detection, ESP32, Python, automated tests and CI

Event-Driven Serverless E-Commerce Backend on AWS 2026
github.com/adeliusa486/PP3-CE-event-driven-ecommerce-backend

- Serverless order system built from API Gateway, Lambda, DynamoDB, and EventBridge. Step Functions roll back failed orders automatically, and Terraform builds the whole stack from code.

Automated CI/CD and GitOps Pipeline on AWS 2026
github.com/adeliusa486/PP4-CE-DV-ecommerce-cicd-pipeline

- GitHub Actions pipeline that runs Python tests, scans the infrastructure code for security issues, and deploys to AWS with no manual steps. Terraform manages every environment.

Certificates of Course Completion

- CCNP Enterprise: Core Networking (v9), Cisco Networking Academy
- CCNP Enterprise: Advanced Routing, Cisco Networking Academy
- Network Security, Cisco Networking Academy
- AWS Certified Solutions Architect – Associate (SAA-C03), 96-hour course, Cantrill.io

Skills and Languages

Research: Experiment design, statistical testing, reproducible research, scientific writing, responses to reviewers
AI and ML: Python, PyTorch, scikit-learn, NumPy, SHAP, federated learning, YOLO object detection
Networking and Security: SDN, Mininet, Open RAN, OSPF, BGP, EIGRP, VLANs, ACLs, Wireshark, SUMO
Cloud and DevOps: AWS, Terraform, Ansible, Docker, GitHub Actions, Linux
Embedded: Raspberry Pi 5, Hailo-8L, ESP32
Languages: Urdu (native), Punjabi (native), English (B2), Arabic (beginner)

References

Dr. Toqeer Ali Syed, Faculty of Computer and Information Systems, Islamic University of Madinah | toqeer@iu.edu.sa